ADJUNCTIONS BETWEEN THE FUNCTORS ON PRESHEAVES ATTACHED TO AN ADJOINT PAIR OF FUNCTORS

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The functors on presheaves attached to a functor of categories. Same chapter, the definitions of the direct and inverse image functors on presheaves and of the auxiliary categories of morphisms and of objects mapping into a fixed object. Supplies the functors and the categories appearing in the statement.

Lemma 0.1. *Let $u : \mathcal{C} \to \mathcal{D}$ and $v : \mathcal{D} \to \mathcal{C}$ be functors of categories. Assume that v is right adjoint to u. Then we have*

(1) $u^p h_V = h_{v(V)}$ *for any V in $\mathcal{D}$,*
(2) *the category $\mathcal{I}^v_U$ has an initial object,*
(3) *the category ${}^u_V\mathcal{I}$ has a final object,*
(4) ${}_pu = v^p$, *and*
(5) $u^p = v_p$.

Proof. Proof of (1). Let V be an object of $\mathcal{D}$. We have $u^p h_V = h_{v(V)}$ because $u^p h_V(U) = \mathrm{Mor}_{\mathcal{D}}(u(U), V) = \mathrm{Mor}_{\mathcal{C}}(U, v(V))$ by assumption.

Proof of (2). Let U be an object of $\mathcal{C}$. Let $\eta : U \to v(u(U))$ be the map adjoint to the map $\mathrm{id} : u(U) \to u(U)$. Then we claim $(u(U), \eta)$ is an initial object of $\mathcal{I}^v_U$. Namely, given an object $(V, \phi : U \to v(V))$ of $\mathcal{I}^v_U$ the morphism ϕ is adjoint to a map $\psi : u(U) \to V$ which then defines a morphism $(u(U), \eta) \to (V, \phi)$.

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Proof of (3). Let V be an object of $\mathcal{D}$. Let $\xi : u(v(V)) \to V$ be the map adjoint to the map $\mathrm{id} : v(V) \to v(V)$. Then we claim $(v(V), \xi)$ is a final object of ${}^u_V\mathcal{I}$. Namely, given an object $(U, \psi : u(U) \to V)$ of ${}^u_V\mathcal{I}$ the morphism ψ is adjoint to a map $\phi : U \to v(V)$ which then defines a morphism $(U, \psi) \to (v(V), \xi)$.

Hence for any presheaf $\mathcal{F}$ on $\mathcal{C}$ we have

$$\begin{aligned} v^p\mathcal{F}(V) &= \mathcal{F}(v(V)) \\ &= \mathrm{Mor}_{PSh(\mathcal{C})}(h_{v(V)}, \mathcal{F}) \\ &= \mathrm{Mor}_{PSh(\mathcal{C})}(u^p h_V, \mathcal{F}) \\ &= \mathrm{Mor}_{PSh(\mathcal{D})}(h_V, {}_pu\mathcal{F}) \\ &= {}_pu\mathcal{F}(V) \end{aligned}$$

which proves part (4). Part (5) follows by the uniqueness of adjoint functors. $\square$